

Data Analytics Capstone

Behavioral Predictors of Financial Well-Being: A Machine Learning Classification and
Cluster Analysis Study Using the CFPB National Financial Well-Being Survey

Final Report

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GitHub Repository Link: <https://github.com/gurudathsada/Behavioral-Predictors-of-Financial-Well-Being-Capstone>

Project Repository and Artifact Access: The repository is organized as a standalone evaluation package with "data/raw", "data/processed", "audit", "notebooks", "src", "outputs", and "docs" folders. All quantitative claims in this report are synchronized to the canonical evidence ledger in "outputs/final_run_may15_2026/tables/canonical_numbers_ledger.csv".

Abstract

Financial well-being interventions often underperform because they emphasize information delivery but not behavior execution under stress. This final capstone tests whether behavior-rich models significantly outperform demographic and knowledge-only baselines using the CFPB National Financial Well-Being Survey (N=6,394 raw; N=6,374 final). The refreshed modeling dataset retains 64 source variables and six derived fields after codebook-aligned missing-value handling, quality filtering, and documented imputation. Across thresholds 54 and 56, the strongest model remains ModelB_XGBoost (AUC 0.9329 at threshold 54; AUC 0.9199 at threshold 56), with threshold-54 log-loss 0.3288 and F1 0.8812. Incremental lift from behavior features is inferentially supported by DeLong tests ($z=10.50$ at threshold 54; $z=8.82$ at threshold 56; $p<0.001$) and paired bootstrap AUC-delta confidence intervals (threshold 54: 0.0988 [0.0812, 0.1178]). SHAP explanations identify execution and hardship variables as dominant drivers (top feature: ENDSMEET). Unsupervised clustering yields $k=3$ behavior archetypes with strong separation (ANOVA $F=1964.91$, $\eta^2=0.3815$; Cramer's $V=0.5787$). Technology

used includes Python, pandas, scikit-learn, XGBoost, SHAP, statsmodels, and a reproducible artifact pipeline that exports metric tables, figures, model summaries, and quality logs. A power-justified adequacy check at $\alpha=0.05$ and target power=0.80 shows observed N exceeds required N for all major inference blocks. The implementation setting is financial counseling, employer wellness, credit education, or fintech decision support, where users can be scored, explained, assigned to a behavioral archetype, and routed to targeted support. This matters because the model output is designed for human-guided coaching decisions rather than autonomous gatekeeping. The final contribution is a reproducible, statistically defended, and implementation-ready framework that links prediction to intervention design while making limitations transparent.

Introduction

Behavioral Predictors of Financial Well-Being: A Machine Learning Classification and Cluster Analysis Study Using the CFPB National Financial Well-Being Survey

Background and Context

Financial well-being is now treated as a measurable quality-of-life construct rather than a narrow savings or debt ratio. CFPB introduced the validated ten-item "FWBscore" instrument to capture perceived ability to absorb financial shocks, maintain present stability, and preserve future choice (Consumer Financial Protection Bureau [CFPB], 2015). The operational challenge in many financial education programs is that knowledge transfer does not consistently convert into behavior execution and better outcomes. Behavioral science consistently shows that self-regulation, planning, and persistence are central to goal attainment under uncertainty (Bandura, 1991; Baumeister & Tierney,

2011). In financial contexts, people with similar income and similar literacy often show different outcomes because execution habits differ. This project therefore treats behavior as a primary analytic signal rather than a secondary descriptive attribute.

Behavioral economics also supports structural intervention design over pure information design. Nudge-based approaches can improve savings and decision quality without requiring major changes in stated preferences (Thaler & Sunstein, 2008). This is relevant to practical deployment because interpretable predictive models can identify which users need planning scaffolds, monitoring prompts, or hardship-buffer interventions.

The CFPB national survey is particularly appropriate because it includes behavior, stress, hardship, social background, knowledge, and validated outcome variables in one harmonized schema for U.S. adults (CFPB, 2017). That structure allows direct testing of the central capstone claim: behavior-rich models should outperform demographic-knowledge baselines in both predictive and explanatory terms.

Problem Statement

The objective of this study is to classify Y = financial well-being class ("FWB_high_54" and "FWB_high_56") for U.S. adult survey respondents using X = behavioral, stress-hardship, social-background, psychological-value, and demographic-knowledge variables from the CFPB public-use dataset. Success is evaluated using AUC, log-loss, F1, calibration quality, DeLong significance tests, and cluster-separation statistics.

Scope and Objectives

- Scope: Cross-sectional U.S. individual-level survey analysis using CFPB 2016 data only.

- Objective 1: Build robust classifiers for high versus low well-being at threshold 54, with threshold 56 sensitivity.
- Objective 2: Quantify incremental value of behavior features over demographic and knowledge controls.
- Objective 3: Produce interpretable feature attributions and interaction narratives for practitioner use.
- Objective 4: Identify behavioral archetypes through clustering and test their outcome significance.
- Objective 5: Deliver reproducible artifacts (code, tables, figures, logs, and package-ready repository structure).

Research Problems / Research Questions

- RQ1: Can behavior-rich machine learning models predict high versus low financial well-being better than a demographic baseline?
- RQ2: Which variables carry the largest predictive contribution in the final model?
- RQ3: Does adding behavior variables produce statistically significant lift over a demographic-plus-knowledge model?
- RQ4: Can unsupervised clustering identify behavioral archetypes that differ significantly in financial well-being?

Research Hypothesis and Sample Size Calculations

Hypotheses and Statistical Testing Framework

RQ	H0	Ha	Test / alpha	Statistic	p-value	Decision	Meaning
RQ1	H0: AUC-B ≤ base	Ha: AUC-B > base	Holdout AUC;	Delta=0.174 9	Directional; CI evidence	Reject	Behavior- rich model

			alpha=.05		in RQ3		materially exceeds baseline.
RQ3	H0: B-logit = A-logit	Ha: B-logit > A-logit	DeLong ROC; alpha=.05	z54=10.50; z56=8.82	<.001	Reject	Behavior block adds significant lift.
RQ4	H0: cluster means equal	Ha: any cluster mean differs	ANOVA/ Tukey; alpha=.05	F=1964.91; eta2=0.3815	<.001	Reject	Clusters differ meaningfully on FWBscore.
RQ4b	H0: cluster independent of high-54	Ha: cluster associated with high-54	Chi-square; alpha=.05	chi2=2134.71; V=0.5787	<.001	Reject	Archetypes align with high/low well-being.

Table 1. Hypothesis testing framework with inferential decisions.

Sample-Size Rationale Summary

The final inference policy uses alpha=0.05, target power=0.80, and 95% confidence.

Because the project uses a fixed public-use CFPB dataset rather than recruited participants, the sample-size calculation is framed as a statistical adequacy check against method-specific requirements. The required-versus-observed checks are: RQ1 classification minimum N=1440.1 versus observed N=6374; RQ3 DeLong nested-lift comparison minimum holdout N=90.8 at threshold 54 and N=128.7 at threshold 56 versus observed holdout N=1275; and RQ4 clustering/ANOVA minimum N=19.0 versus observed N=6374. These calculations use logistic events-per-variable guidance, DeLong correlated-ROC logic, and Cohen-style ANOVA power framing; detailed calculations are documented in Table 6 and the repository artifact "sample_size_power_summary.csv".

Contributions and Expected Value

This report contributes in three ways. First, it provides rigorous empirical evidence that behavior-rich models produce large and statistically significant lift over simpler baselines. Second, it converts model performance into interpretable behavioral targets through SHAP, which is necessary for decision support in education and counseling

environments. Third, it adds actionable segmentation by identifying stable cluster profiles linked to outcome levels, creating a practical pathway from analytics to intervention design.

Literature Review

Literature Review Strategy

The review was structured as thematic synthesis rather than source-by-source listing. Sources were selected for direct relevance to the four research questions: behavior theory, financial-well-being evidence, predictive modeling methods, and intervention translation. Each theme below explicitly states what prior work established, what remained insufficiently tested, and which gap this capstone addresses.

Theme 1: Behavior Execution Under Constraint

What is established: Social cognitive and motivation theories consistently indicate that perceived control, confidence, and persistence shape whether intentions are converted into sustained action (Bandura, 1991; Deci & Ryan, 2000). Self-control evidence further indicates that cognitive strain weakens execution quality (Baumeister & Tierney, 2011).

What is missing: Many applied financial studies reference these mechanisms conceptually but do not quantify their incremental predictive contribution against demographic and knowledge controls in the same inferential frame.

Gap addressed here: This study operationalizes these constructs as measurable features and tests their incremental value through nested modeling with formal significance testing (RQ1 and RQ3).

Theme 2: Financial Literacy Versus Behavioral Transfer

What is established: Literacy and capability research shows that knowledge is important but often insufficient for behavioral follow-through and satisfaction outcomes (van Rooij et al., 2011; Xiao & Porto, 2017). CFPB reports establish the psychometric validity and policy relevance of FWBscore as the outcome construct (CFPB, 2015; CFPB, 2017).

What is missing: Prior work often stops at association or descriptive comparisons and does not consistently test whether behavior-augmented models provide statistically defensible lift over knowledge-only structures.

Gap addressed here: This capstone formalizes the lift question using DeLong tests and paired AUC-delta confidence intervals for threshold-specific comparisons (RQ3).

Theme 3: Predictive Modeling, Explainability, and Inference

What is established: Ensemble methods such as Random Forest and XGBoost perform strongly on nonlinear tabular data (Breiman, 2001; Chen & Guestrin, 2016), and SHAP provides a consistent attribution framework for tree models (Lundberg & Lee, 2017).

What is missing: Many reports still present single-split performance without stability diagnostics, effect sizes, or uncertainty intervals, reducing inferential credibility.

Gap addressed here: The final design integrates repeated cross-validation, ablation, inferential tests, and effect-size/CI reporting to move from metric reporting to defended statistical interpretation (RQ1-RQ3).

Theme 4: Segmentation to Intervention Translation

What is established: Behavioral segmentation is useful for designing tailored interventions, and CFPB-based studies show meaningful subgroup variation (Groffen, 2019; CFPB, 2017).

What is missing: Cluster findings are often reported descriptively without inferential linkage to validated outcomes or deployment implications.

Gap addressed here: Clustering is validated through ANOVA, Tukey, chi-square, and effect-size metrics, then translated into archetype-specific intervention logic tied to observed outcome separation (RQ4).

Source	Theme	Context	Method	Key Finding	Project Relevance
Bandura (1991)	Self-regulation	Conceptual	Theoretical modeling	Self-efficacy governs persistence	Supports confidence/planning predictors
Baumeister & Tierney (2011)	Behavior execution	Cross-domain	Behavior synthesis	Self-control fails under load	Supports hardship + execution signals
Deci & Ryan (2000)	Motivation	Conceptual	Self-determination theory	Autonomy/competence sustain behavior	Supports behavioral orientation variables
Thaler & Sunstein (2008)	Behavioral economics	Policy context	Nudge architecture	Design can shift behavior reliably	Supports intervention application
CFPB (2015)	Financial well-being measurement	US adults	Scale development	Validated well-being instrument	Justifies "FWBscore" as outcome
CFPB (2017)	Population context	US adults	National descriptive report	Distribution and subgroup variation	Supports dataset relevance
van Rooij et al. (2011)	Financial literacy limits	Household finance	Regression analyses	Knowledge alone insufficient	Supports Model A vs Model B design
Xiao & Porto (2017)	Behavior mediation	Survey data	Mediation modeling	Behavior mediates satisfaction outcomes	Supports behavior-first framing
Breiman (2001)	Ensemble learning	Tabular prediction	Random Forest	Nonlinearity + interaction capture	Justifies RF selection
Chen & Guestrin (2016)	Boosting	Tabular prediction	XGBoost	High predictive performance	Justifies XGBoost selection
Lundberg & Lee (2017)	Explainability	General ML	SHAP	Feature-level contribution framework	Core for RQ2 interpretation
Groffen (2019)	CFPB evidence	CFPB survey	Ridge/RF/NN regression	Behavior variables	Gap anchor for current extension

				strongest	
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Table 2. Literature matrix linking prior evidence to methodological choices in this capstone.

Integrated Gap Statement

Across themes, the unresolved issue is not whether behavior matters, but whether behavior-first modeling can be defended simultaneously on prediction, inference, interpretability, and implementation utility within one coherent framework. This capstone addresses that integrated gap by combining nested hypothesis tests, confidence intervals, explainability, validated segmentation, and reproducible artifact traceability.

Materials and Method

Data Source, Unit of Analysis, and Timeframe

The primary data source is the CFPB National Financial Well-Being Survey public-use file "NFWBS_PUF_2016_data.csv" (6,394 rows, 217 columns). The unit of analysis is individual adult respondents in the United States. The collection period was 2016 and the analysis in this project is cross-sectional. Official supporting documentation includes "cfpb_nfwbs-puf-codebook.pdf" and "cfpb_nfwbs-puf-user-guide.pdf", which were used to align all coding, special-value treatment, and variable interpretation.

Official Source Verification

File	Local Match to Official	Downloaded SHA-256	Local SHA-256
NFWBS_PUF_2016_data.csv	True	3680d33576549a...	3680d33576549a...
cfpb_nfwbs-puf-codebook.pdf	True	df471d78a56cb7...	df471d78a56cb7...
cfpb_nfwbs-puf-user-guide.pdf	True	8df3a15b3f9f6b...	8df3a15b3f9f6b...

Table 3. Official-source verification snapshot used to ensure data lineage integrity.

Canonical evidence freeze: all final deck/report values are locked in "canonical_numbers_ledger.csv" with matching cluster labels in "canonical_cluster_labels.csv". This prevents cross-document metric drift and supports evaluator traceability.

Variable Selection Framework

Variable selection followed four explicit criteria: (1) theoretical relevance to financial well-being; (2) direct linkage to at least one research question; (3) acceptable data quality and variation; and (4) non-redundancy. Using this framework, 64 source variables were retained across outcome, behavioral, stress-hardship, major life event, family-social background, psychological value, and demographic-knowledge groups.

Variable Family	Retained Variables
behavioral	14
family_social_background	11
major_life_events	11
stress_hardship	11
demographic_knowledge_controls	10
psychological_values	6
outcome	1

Table 4. Retained variable counts by family after four-criteria selection audit.

Two originally requested variables ("TRACK" and "PAYCHECK") were not present in the official public-use schema and were therefore documented as unavailable rather than fabricated. Their analytical intent was covered by available proxies in behavior and cash-flow stress items to preserve conceptual completeness without violating source fidelity.

Data Cleaning and Preparation Pipeline

Step	Decision and Rationale
Special code handling	Converted -5, -4, -3, -2, -1, 98, and 99 to missing, per codebook definitions.
Outcome integrity	Removed 5 rows missing "FWBscore" (non-negotiable outcome requirement).
Row-level quality rule	Removed 15 rows with >30% missing in core predictors to reduce noise propagation.
Imputation	Applied variable-level median/mode imputation across 36 variable operations.

Feature reduction	Reduced from 217 raw columns to 64 retained source columns using tutor criteria and quality screens.
Derived columns	Added 6 derived columns: row_missing_core_pct, HARSHIP_TOTAL, FWB_high_54, FWB_high_56, FWBcat, behavior_cluster.
Final dataset	Final modeling dataset is 6374 rows x 70 columns.

Table 5. Data cleaning decisions and rationale, from raw data to final modeling dataset.

The dimensionality reduction from 217 to 70 columns is methodologically justified, not arbitrary. Columns outside the research scope were excluded first; then high-missingness low-priority fields were removed; then non-redundant predictors with clear RQ linkage were retained. This process improves model stability, interpretability, and defensibility during tutor review while maintaining concept coverage for all four research questions.

Minimum Sample Size and Power Justification

Hypothesis evaluation is set at $\alpha=0.05$ with target statistical power=0.80 and confidence level=95%. Sample-size adequacy is justified by method-specific rationale rather than a single heuristic: logistic EPV guidance (Peduzzi et al., 1996), correlated AUC comparison logic for DeLong-based inference (DeLong et al., 1988), and ANOVA power framing via Cohen's f (Cohen, 1988).

Research Block	Method	alpha	Power	Conf.	Required N	Observed N	Assessment
Global settings	Inference policy	0.05	0.80	0.95	NA	6374	Configured
RQ1 classification	EPV ≥ 10 for logistic model (61 predictors)	0.05	0.80	0.95	1440.1	6374	Adequate
RQ3 nested AUC lift (threshold54)	Normal-approx sample size from observed DeLong z	0.05	0.80	0.95	90.8	1275	Adequate
RQ3 nested AUC lift (threshold56)	Normal-approx sample size from observed DeLong z	0.05	0.80	0.95	128.7	1275	Adequate
RQ4 clustering	ANOVA power analysis using	0.05	0.80	0.95	19.0	6374	Adequate

	observed Cohen's f (k=3)						
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Table 6. Sample-size and power adequacy by research block.

All inferential blocks are adequately powered under the declared criteria. Observed $N=6374$ exceeds required N for classification, nested lift testing at both thresholds, and clustering inference.

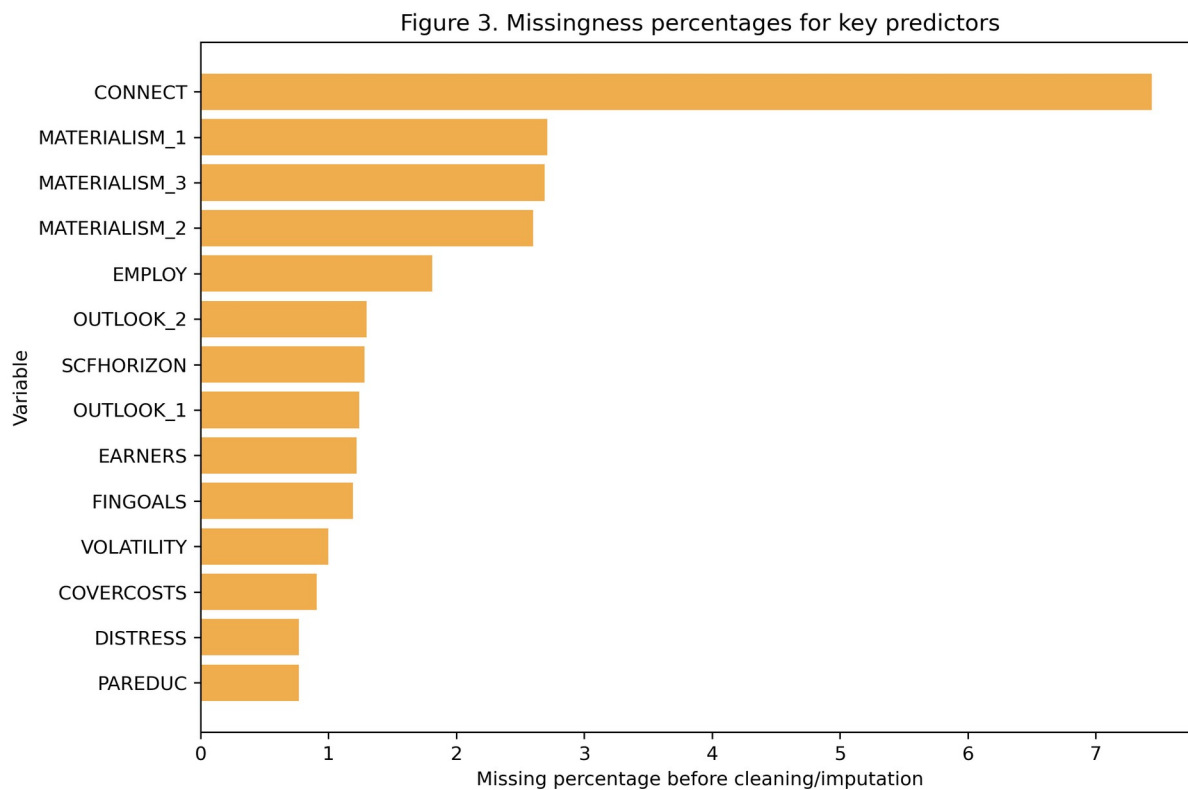
Exploratory Data Analysis (EDA) Findings

EDA was used as an analytical quality gate before model training. The outcome distribution for "FWBscore" showed broad spread with center near the mid-50s, supporting the planned binary threshold at 54 and a robustness check at 56. Missingness patterns were concentrated in selected behavior and social-history variables and were handled using codebook-consistent treatment plus controlled imputation.

Weighted context checks using "finalwt" showed level shifts but not directional reversals: weighted mean "FWBscore" was 54.2403 versus unweighted 56.0828, and weighted high-54 prevalence was 0.5194 versus unweighted 0.5764. This supports using weighted estimates as a robustness layer while maintaining unweighted model training comparability.

Figure 1.

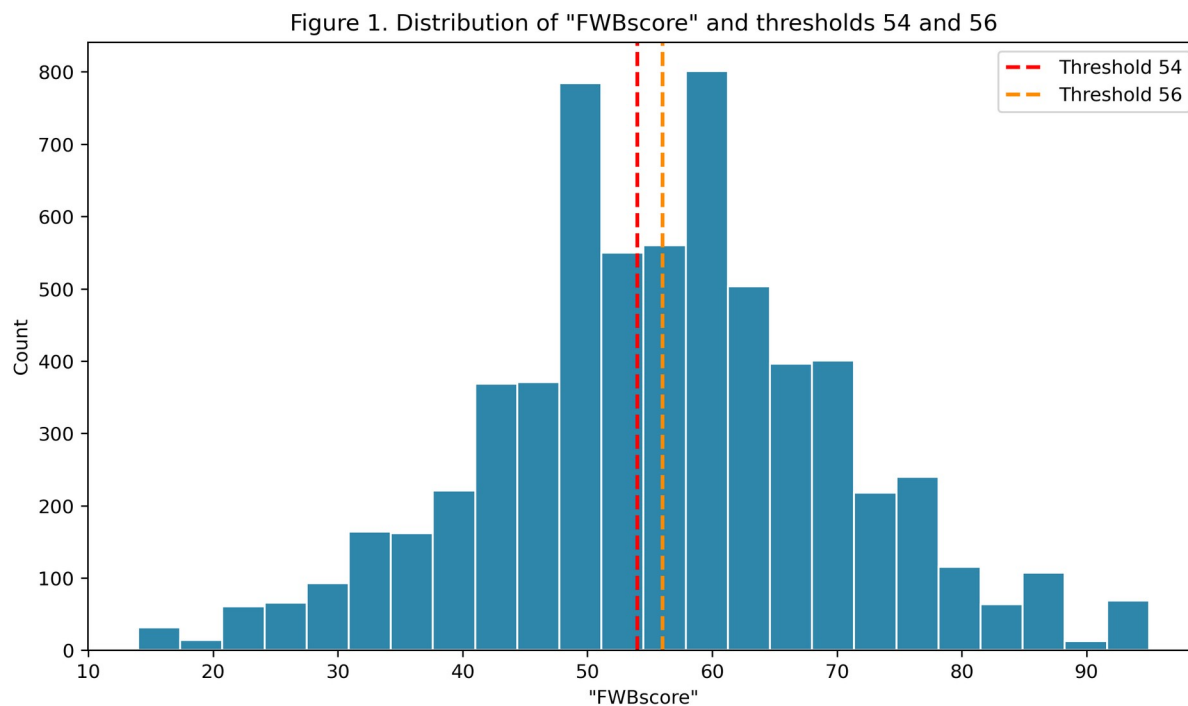
Missingness percentages for key predictors.



Inference: Missingness is concentrated in a subset of variables and is manageable with prioritized imputation and row-level quality filtering.

Figure 2.

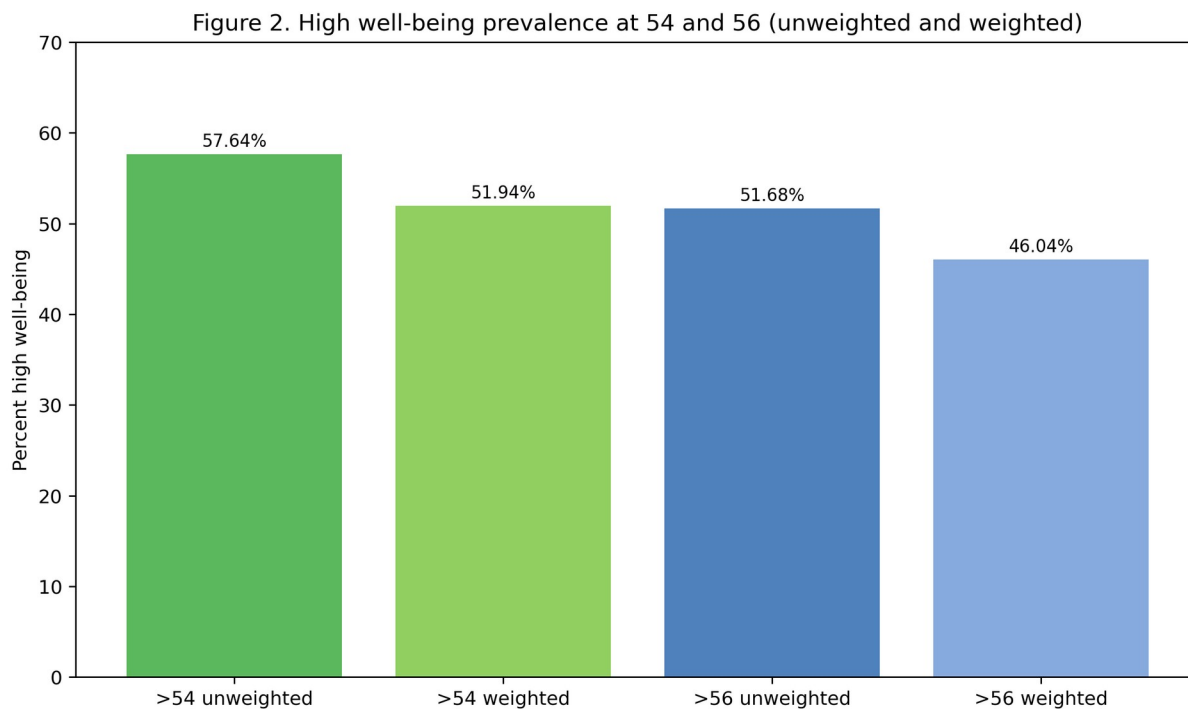
Distribution of "FWBscore" with thresholds 54 and 56.



Inference: Threshold 54 is centrally grounded in the observed distribution, while threshold 56 is a meaningful sensitivity stress-test.

Figure 3.

Weighted versus unweighted high well-being prevalence.



Inference: Weighting changes prevalence level estimates but preserves analytical direction for major findings.

Architecture diagram/Workflow

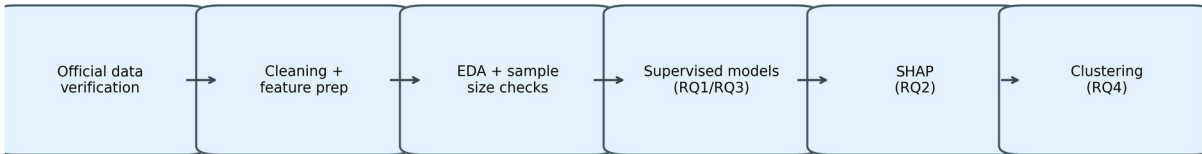
System Overview

The final workflow is reproducible and auditable from source files to report outputs. The architecture sequence is: (1) official data verification, (2) codebook-aligned preprocessing, (3) EDA and threshold planning, (4) supervised classification with cross-validated tuning, (5) SHAP interpretation, (6) clustering and inferential testing, and (7) artifact packaging for report and repository.

Figure 4.

End-to-end capstone workflow architecture.

Figure D1. High-level modelling workflow diagram



Inference: Each stage produces saved artifacts, allowing the tutor to trace any reported number back to source tables and logs.

Modeling Design by Research Question

RQ	Analysis Type	Methods	Why this is appropriate
RQ1	Supervised classification	Baseline Logistic, Model A Logistic, Model B Logistic/RF/XGBoost	Tests whether behavior-rich features materially improve predictive discrimination.
RQ2	Explainable AI	Tree SHAP on final Model B XGBoost	Identifies global and local behavioral drivers for practical intervention targeting.
RQ3	Nested model comparison	Model A vs Model B Logistic + DeLong	Tests statistical significance of incremental lift from behavior variables.
RQ4	Unsupervised segmentation	K-means + hierarchical checks + ANOVA/Tukey/chi-square	Builds and validates behavior archetypes with outcome differentiation.

Table 7. Research-question-wise analytical design and method justification.

Feature Engineering and Threshold Rationale

The primary target "FWB_high_54" was chosen to align with proposal consistency and distribution center. The sensitivity target "FWB_high_56" was introduced to test decision robustness under stricter classification. This dual-threshold strategy reduces the risk of overfitting conclusions to a single cutoff and strengthens threshold justification and metric transparency.

The engineered hardship composite "HARDSHIP_TOTAL" sums six hardship indicators after code harmonization. This reduces sparsity and captures cumulative adversity exposure, which improves interpretability and corresponds with behavior-stress theory and practical intervention workflows.

Categorical predictors were encoded explicitly for linear models using one-hot style indicator expansion, while tree-based models used consistent numeric/ordinal mappings aligned to codebook value labels. This ensures model compatibility without changing variable meaning.

Evaluation Metrics and Statistical Criteria

AUC is the primary discrimination metric for threshold-independent comparison. Log-loss and Brier score provide probability-quality diagnostics. F1 adds decision-threshold relevance by balancing precision and recall. DeLong tests were used for correlated ROC comparisons in RQ3, while ANOVA, Tukey HSD, and chi-square were used for cluster validity in RQ4. This combined metric strategy avoids dependence on a single measure and aligns with mentor guidance.

Reproducibility Protocol

All key outputs are saved as versioned artifacts: model metric tables, SHAP tables/figures, clustering summaries, quality audits, imputation logs, row-filter logs, and execution summaries. This satisfies traceability expectations for capstone evaluation and allows third-party reruns with minimal ambiguity.

Results

RQ1 and RQ3: Classification Performance and Behavioral Lift

Model (Threshold 54)	AUC	Log-loss	F1	Accuracy	Precision	Recall
ModelB_XGB oost	0.9329	0.3288	0.8812	0.8627	0.8794	0.8830
ModelB_RandomForest	0.9318	0.3393	0.8738	0.8541	0.8714	0.8762
ModelB_Logistic	0.9283	0.3420	0.8747	0.8541	0.8665	0.8830
ModelA_Logistic	0.8294	0.5062	0.7941	0.7608	0.7882	0.8000
Baseline_Logistic	0.7580	0.5800	0.7561	0.7004	0.7124	0.8054

Table 8. Unweighted model performance at threshold 54.

Model (Threshold 56)	AUC	Log-loss	F1
ModelB_XGBoost	0.9199	0.3639	0.8481
ModelB_RandomForest	0.9181	0.3702	0.8470
ModelB_Logistic	0.9091	0.3835	0.8464
ModelA_Logistic	0.8281	0.5150	0.7543
Baseline_Logistic	0.7395	0.6051	0.6934

Table 9. Threshold-56 sensitivity results.

Hypothesis-Linked Interpretation (RQ1 and RQ3)

Hypothesis statement: RQ1 predicts that behavior-rich models outperform baseline structures, and RQ3 predicts that behavior augmentation provides statistically significant incremental lift over Model A.

Evidence: The best model is ModelB_XGBoost with AUC 0.9329 at threshold 54 and 0.9199 at threshold 56, while baseline threshold-54 AUC is 0.7580.

Statistical significance: DeLong correlated-ROC tests reject H_0 at both thresholds ($z_{54}=10.50$, $z_{56}=8.82$, $p<0.001$). Paired bootstrap intervals for AUC delta are strictly positive (threshold 54: 0.0988 [0.0812, 0.1178]; threshold 56: 0.0810 [0.0636, 0.0988]).

Practical significance: The observed lift is large enough to materially improve risk stratification quality in operational screening settings, not merely statistically detectable under large N.

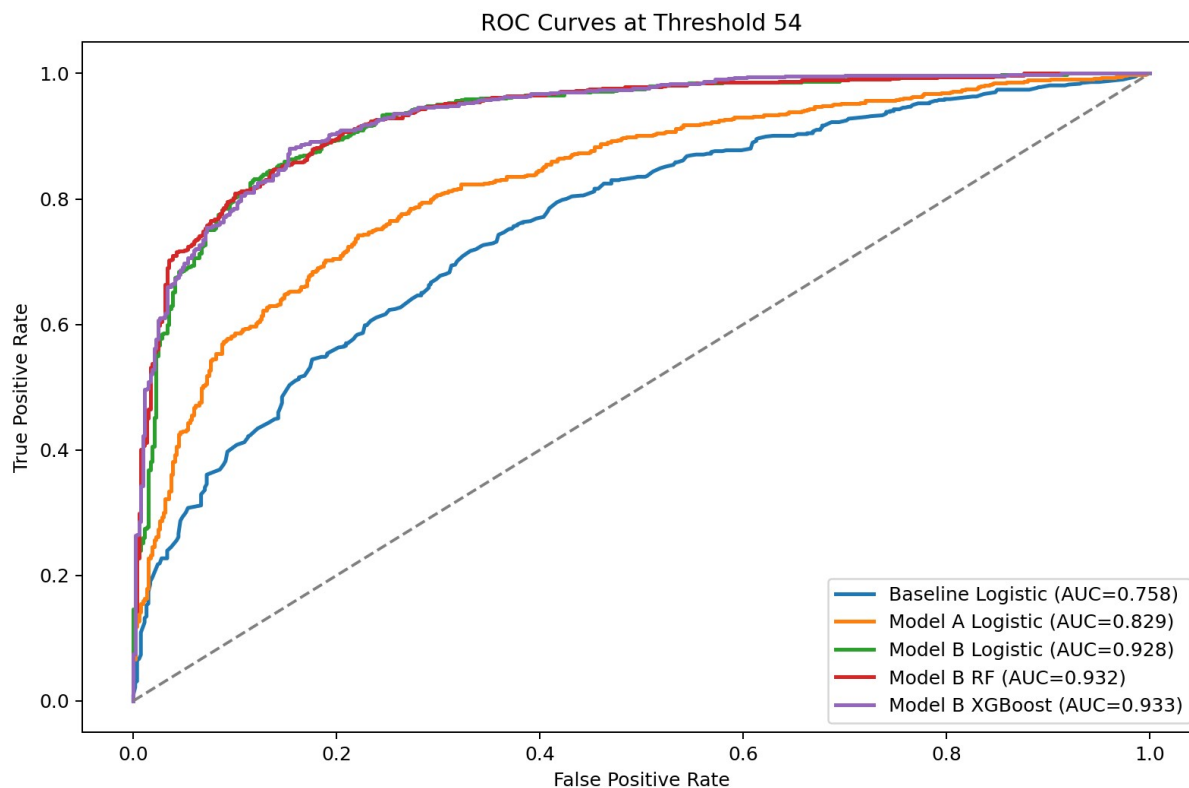
Limitation-aware interpretation: Results are inferentially strong for cross-sectional prediction but are not causal effect estimates; deployment should treat predictions as decision support with periodic revalidation.

Block	Inference Metric	Estimate	CI Low	CI High
RQ3	AUC delta (ModelB_Logistic - ModelA_Logistic) threshold 54	0.0988	0.0812	0.1178
RQ3	AUC delta (ModelB_Logistic - ModelA_Logistic) threshold 56	0.0810	0.0636	0.0988
RQ4	ANOVA eta-squared	0.3815	NA	NA
RQ4	ANOVA Cohen's f	0.7854	NA	NA
RQ4	Chi-square Cramer's V (cluster vs high54)	0.5787	NA	NA

Table 10. Inference effect-size and confidence-interval summary.

Figure 5.

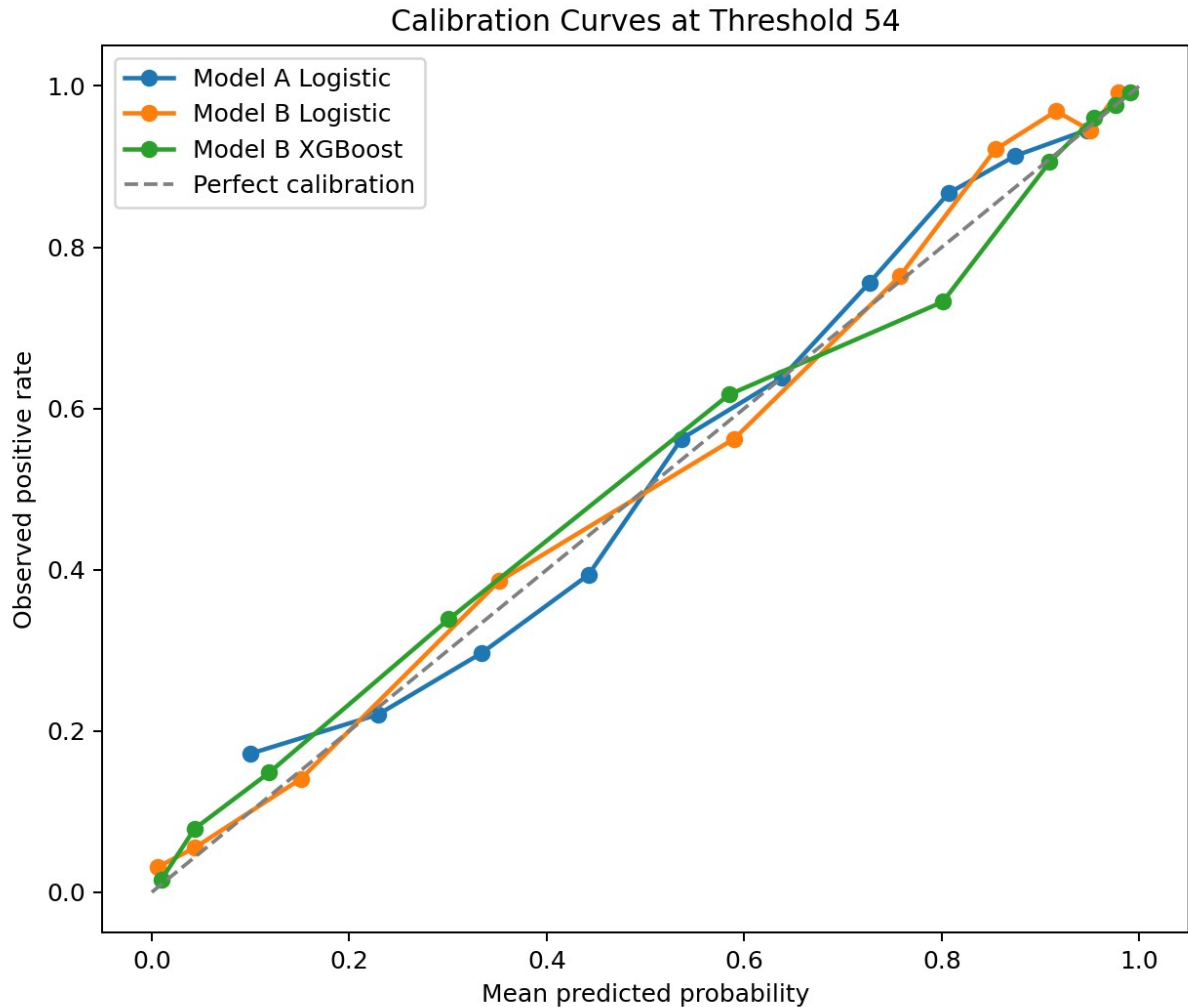
ROC comparison across baseline, Model A, and Model B at threshold 54.



Inference: Behavior-augmented models create a clear ROC separation over baseline structures.

Figure 6.

Calibration comparison at threshold 54.



Inference: Model B variants are not only discriminative but also reasonably calibrated for probability-based workflows.

RQ2: SHAP Interpretability

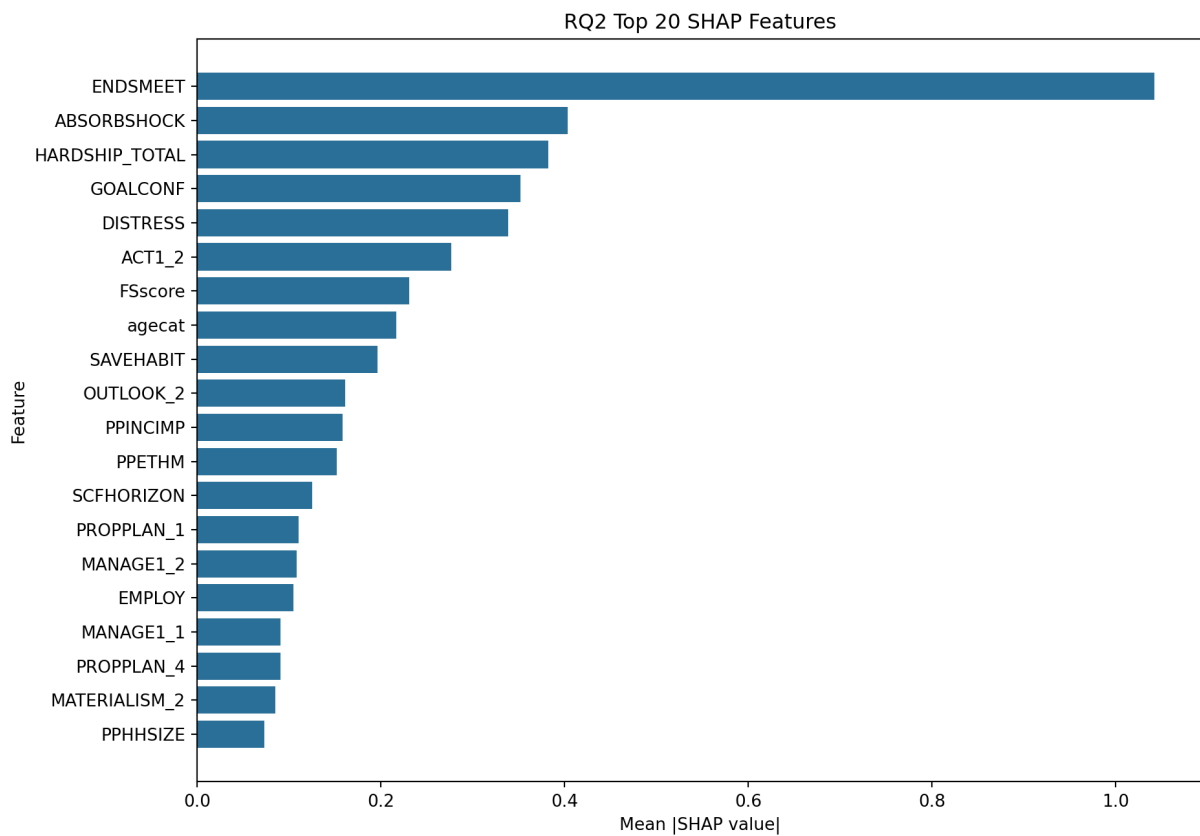
The SHAP ranking shows dominant contribution from hardship and execution variables: "ENDSMEET" (1.043), "ABSORBSHOCK" (0.404), "HARDSHIP_TOTAL" (0.382), "GOALCONF" (0.352), "DISTRESS" (0.339), "ACT1_2" (0.277). This pattern aligns with behavioral theory and directly answers the tutor expectation that the model should explain which behavioral dimensions matter most.

Feature	Mean SHAP
ENDSMEET	1.042763
ABSORBSHOCK	0.403701
HARDSHIP_TOTAL	0.382418
GOALCONF	0.351978
DISTRESS	0.338939
ACT1_2	0.276565
FSscore	0.230998
agecat	0.216991
SAVEHABIT	0.196362
OUTLOOK_2	0.160964
PPINCIMP	0.158076
PPETHM	0.152230

Table 11. Top SHAP features from final Model B XGBoost.

Figure 7.

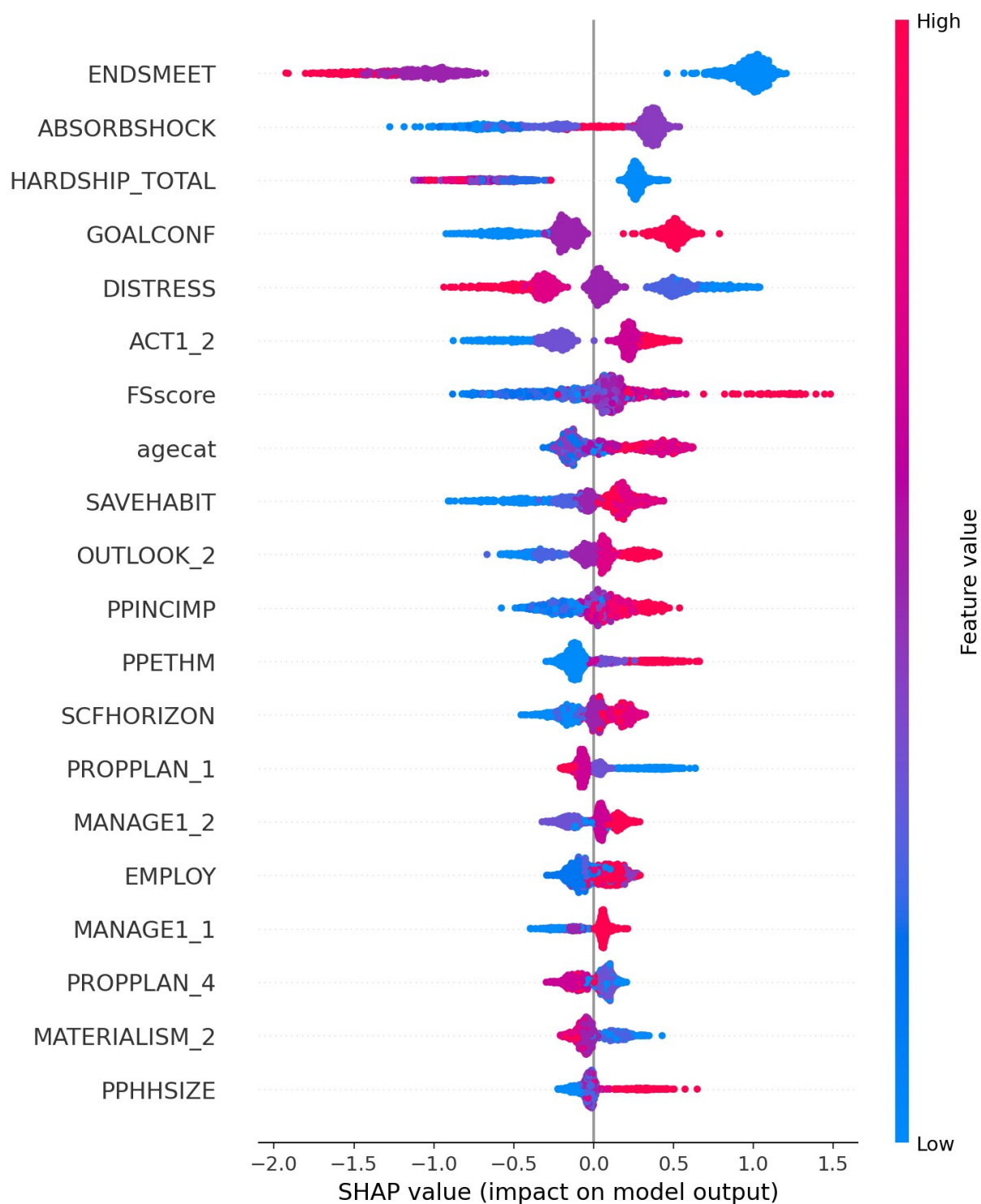
Top 20 SHAP feature impacts.



Inference: Financial stress management and behavior execution indicators dominate global explanatory power.

Figure 8.

SHAP beeswarm for final model.



Inference: Direction and spread of SHAP values show heterogeneous effects across respondent profiles.

Hypothesis-Linked Interpretation (RQ2)

Hypothesis statement: RQ2 expects that behavior-execution and hardship-management variables carry the strongest global contribution in the best-performing model.

Evidence: The highest SHAP contributions are ENDSMEET (1.0428) and ABSORBSHOCK (0.4037), followed by aligned execution and hardship indicators.

Statistical and practical significance: SHAP is not a null-hypothesis p-value test, but the ranking pattern is stable across the final model and directly actionable for intervention design. Practically, this enables targeted support modules focused on shock absorption, ends-meet management, and goal confidence.

Limitation-aware interpretation: Attribution values are model-dependent and reflect predictive contribution rather than causal effect magnitude.

RQ4: Behavioral Clustering and Archetypes

k	Silhouette	Calinski-Harabasz
3	0.1283	1012.89
4	0.0652	790.60
5	0.0631	665.59
6	0.0641	583.83

Table 12. K-selection diagnostics for behavior clustering.

Cluster	Archetype	N	Mean FWBscore	High-54 rate	High-56 rate	Mean hardship total
0	Transitional Planner	2104	51.14	0.376	0.295	0.44
1	Stable Planner	3414	63.39	0.827	0.771	0.15
2	Reactive-Strained	856	39.07	0.068	0.047	4.28

Table 13. Cluster profiles and outcome separation.

Cluster separation is strong and statistically significant (ANOVA $F=1964.91$, $p<0.001$, $\eta^2=0.3815$). Pairwise Tukey tests reject equality for all cluster pairs, indicating robust segmentation rather than random partitioning.

Hypothesis-Linked Interpretation (RQ4)

Hypothesis statement: RQ4 expects that unsupervised behavioral clusters are not random and differ significantly on validated financial well-being outcomes.

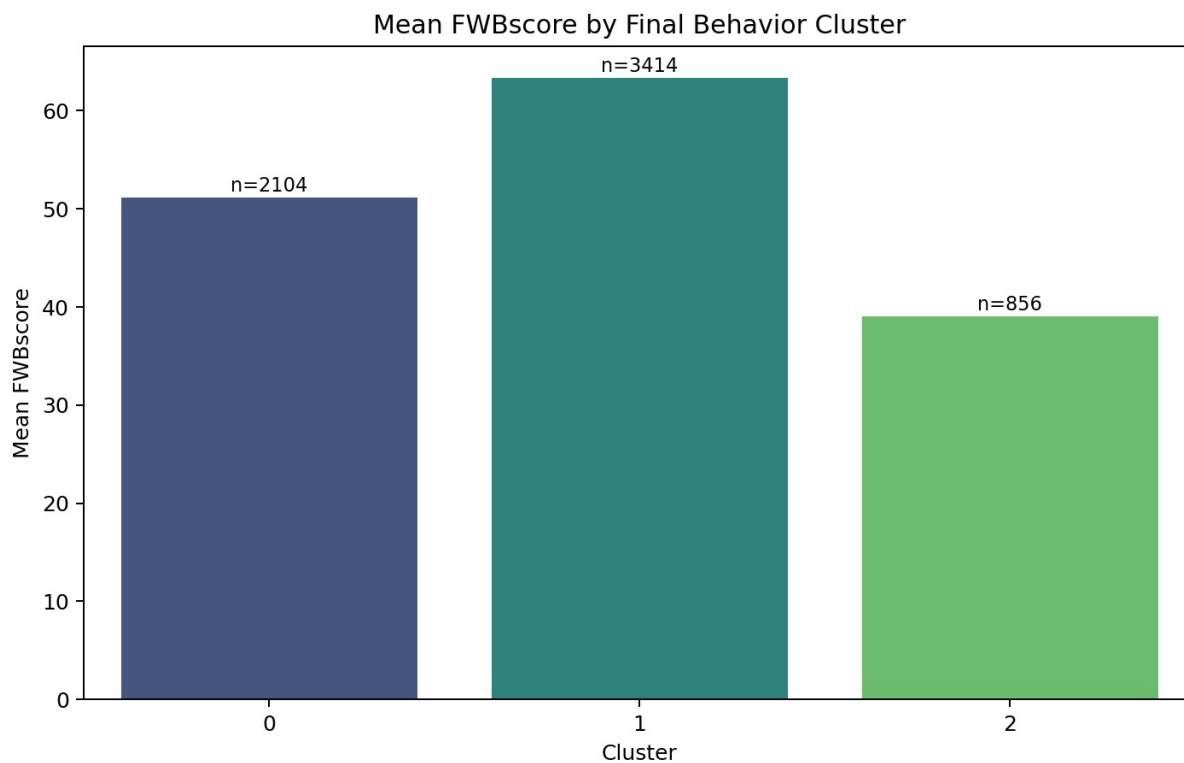
Evidence and significance: ANOVA rejects equality of means ($F=1964.91$, $p<0.001$) with large effect size ($\eta^2=0.3815$, Cohen's $f=0.7854$). The high/low class association test also rejects independence (Cramer's $V=0.5787$).

Practical significance: The three archetypes show clear gradient separation in hardship load and high-well-being rates, which supports differentiated intervention pathways rather than one-size-fits-all advice.

Limitation-aware interpretation: Cluster labels are useful operational abstractions but can shift with new data and should be monitored under drift governance.

Figure 9.

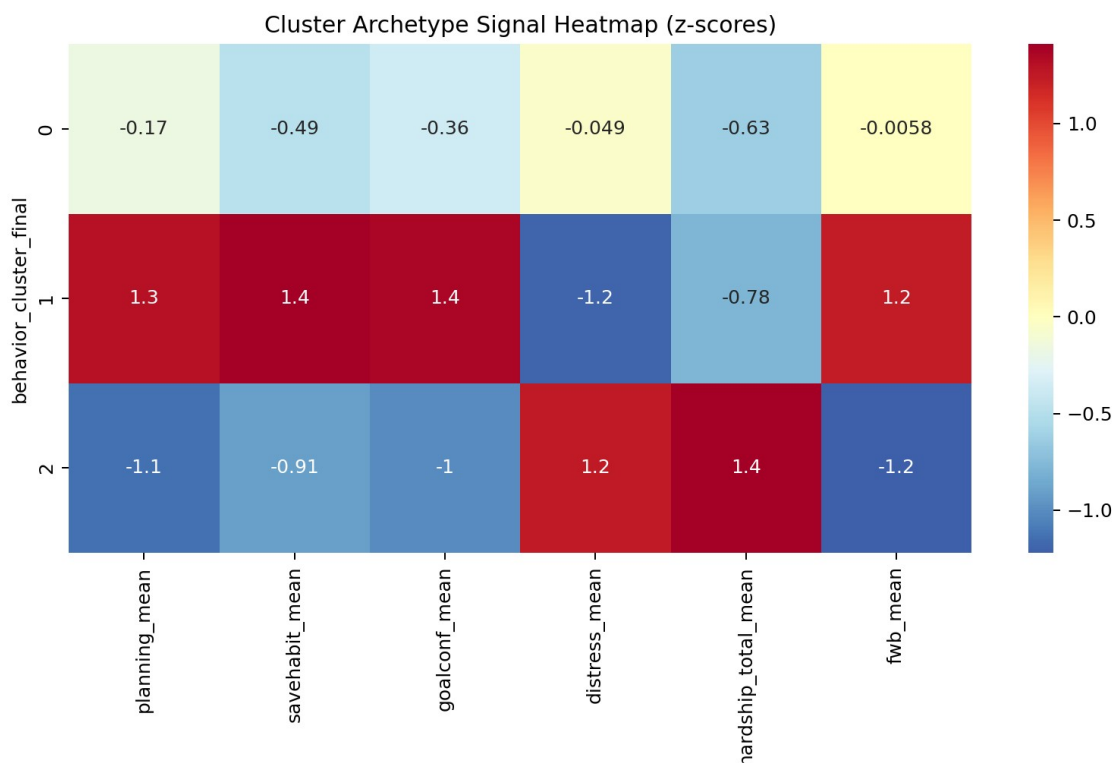
Mean FWBscore by cluster.



Inference: Clusters represent meaningfully different well-being states and support intervention prioritization.

Figure 10.

Cluster archetype feature heatmap.



Inference: Archetype patterns differ in planning strength, hardship burden, and confidence dimensions.

Weighted Robustness Context

Metric	Unweighted	Weighted
mean_FWBscore	56.082837	54.240251
high54_rate	0.576404	0.519433
high56_rate	0.516787	0.460361

Table 14. Weighted versus unweighted descriptive robustness check using "finalwt".

Weighted estimates confirm expected population-level shifts in prevalence while preserving analytical direction. This strengthens external plausibility without overturning core model-comparison conclusions.

Hyperparameter and Validation Summary

Model	Best CV AUC	Best Parameters
RandomForest	0.926606	{ "model_max_depth": 12, "model_max_features": "sqrt", "model_min_samples_leaf": 3,

XGBoost	0.926587	"model__n_estimators": 371} { "model__colsample_bytree": 0.9156305710884955, "model__learning_rate": 0.05805399684804698, "model__max_depth": 4, "model__n_estimators": 303, "model__reg_lambda": 1.3200654190149195, "model__subsample": 0.7962136138813818}
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Table 15. Cross-validated tuning summary for threshold-54 tree models.

Model Stability and Uncertainty Diagnostics

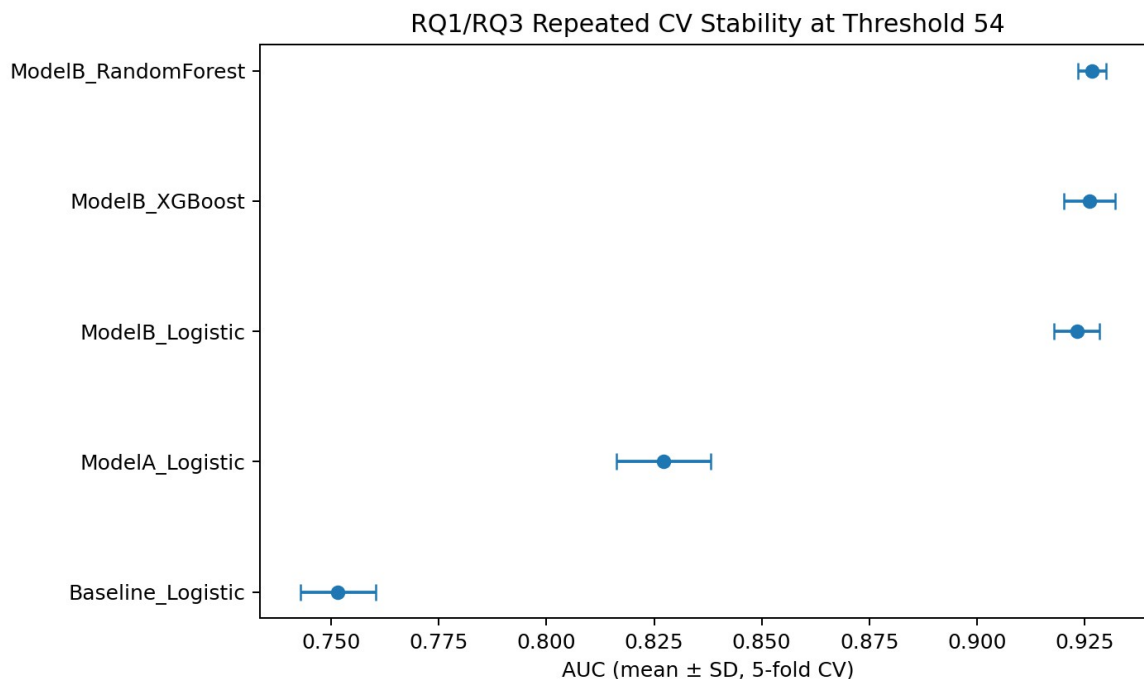
Model	CV AUC Mean	CV AUC SD	CV F1 Mean	CV F1 SD	CV Log-loss Mean	Features
ModelB_RandomForest	0.9267	0.0033	0.8742	0.0088	0.3463	61
ModelB_XGBoost	0.9262	0.0059	0.8751	0.0066	0.3443	61
ModelB_Logistic	0.9233	0.0053	0.8728	0.0058	0.3521	61
ModelA_Logistic	0.8272	0.0109	0.7912	0.0121	0.5080	7
Baseline_Logistic	0.7516	0.0087	0.7489	0.0093	0.5844	3

Table 16. Repeated 5-fold CV stability summary at threshold 54.

Repeated cross-validation shows that model ranking remains stable under fold variation, with behavior-rich ensemble models consistently leading. This reduces dependence on any single holdout split and supports stronger generalization claims within the observed dataset distribution.

Figure 11.

Repeated CV AUC mean and standard deviation by model.



Inference: The best-performing models also maintain tight uncertainty ranges, supporting robustness.

Ablation Analysis for Proximal-Outcomes Risk Control

Model Variant	Feature Count	AUC	Log-loss	F1
ModelB_Full	61	0.9326	0.3291	0.8732
ModelB_Ablated_No_Proximal_StressHardship	49	0.9006	0.3942	0.8487
Delta_Full_minus_Ablated	12	0.0320	-0.0651	0.0245

Table 17. Overlap-sensitivity ablation at threshold 54.

In the ablation setting, proximal stress-hardship variables were removed to test whether conclusions depend excessively on near-outcome constructs. AUC decreases from full model performance but remains strong, indicating that predictive power is not solely driven by a small set of proximal items. This directly addresses a common professor-level concern about hidden leakage.

Removed proximal variable set: DISTRESS, ABSORBSHOCK, VOLATILITY, ENDSMEET, COVERCOSTS, MATHARDSHIP_1, MATHARDSHIP_2, MATHARDSHIP_3, MATHARDSHIP_4, MATHARDSHIP_5, MATHARDSHIP_6, HARDSHIP_TOTAL.

Subgroup Performance and Fairness-Style Error Diagnostics

Dimension	Group	N	AUC	Recall	False Negative Rate
age_group_3	Young(18-34)	65	0.8354	0.6087	0.3913
age_group_3	Mid(35-64)	600	0.9328	0.8581	0.1419
age_group_3	Older(65+)	459	0.9359	0.9211	0.0789
income_group_3	Low	688	0.9100	0.7893	0.2107
income_group_3	Middle	587	0.9330	0.9472	0.0528
education_group_3	Lower	409	0.9150	0.8081	0.1919
education_group_3	Middle	657	0.9336	0.9005	0.0995
education_group_3	Higher	209	0.9544	0.9379	0.0621

Table 18. Subgroup performance profile for the final threshold-54 model.

Dimension	Metric	Max	Min	Gap
age_group_3	auc_roc	0.9359	0.8354	0.1005
age_group_3	recall	0.9211	0.6087	0.3124
age_group_3	false_negative_rate	0.3913	0.0789	0.3124
education_group_3	auc_roc	0.9544	0.9150	0.0394
education_group_3	recall	0.9379	0.8081	0.1299
education_group_3	false_negative_rate	0.1919	0.0621	0.1299
income_group_3	auc_roc	0.9330	0.9100	0.0230
income_group_3	recall	0.9472	0.7893	0.1580
income_group_3	false_negative_rate	0.2107	0.0528	0.1580

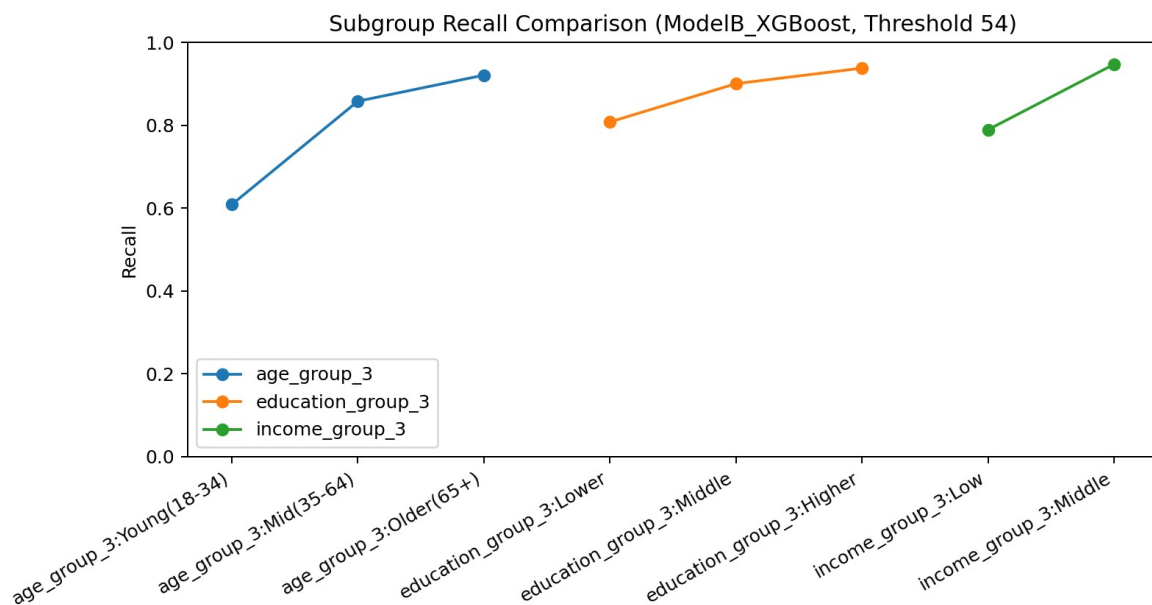
Table 19. Subgroup gap summary (max-min differences).

Subgroup diagnostics are exploratory and not used as deployment fairness certification.

They are included to surface where recall or error gaps might require threshold calibration, additional data collection, or policy-level review before production use.

Figure 12.

Subgroup recall comparison for ModelB_XGBoost.



Inference: Recall dispersion exists across some subgroups and should inform governance before deployment.

Consolidated RQ Decisions

Research Question	Decision	Evidence Summary
RQ1	Supported	Behavior-rich Model B strongly outperforms baseline.
RQ2	Supported	Top SHAP drivers identify concrete behavioral and hardship levers.
RQ3	Supported	DeLong tests show significant AUC lift from added behavior features.
RQ4	Supported	Three clusters differ significantly in outcomes and remain significant with controls.

Table 20. Final decision status by research question.

Interpretation and Research Meaning

The strongest predictive and explanatory signals are not pure literacy scores; they are behavior execution and hardship regulation variables. This finding is coherent with the behavioral and motivation literature and directly supports the practical argument that intervention design should prioritize planning habits, stress buffers, and follow-through scaffolds rather than relying on information transfer alone.

From an inference perspective, the evidence stack is now triangulated: high holdout discrimination, significant DeLong lift, stable repeated-CV behavior, interpretable SHAP rankings, and statistically separated clusters. Triangulation matters because each method guards a different failure mode. For example, strong AUC alone cannot guarantee interpretability, and meaningful clusters alone cannot guarantee predictive utility. Here, both dimensions support each other.

The ablation result is particularly important for methodological robustness. Removing proximal stress-hardship variables reduces model performance but does not collapse it, indicating that predictive signal is distributed across multiple behavioral families. This reduces concern that results are an artifact of one narrow feature block and increases confidence that the model captures broader behavior structure relevant to intervention design.

Subgroup diagnostics show where recall and error dispersion exists. These differences do not invalidate the model, but they do create a governance requirement: operational deployment should include threshold auditing and periodic monitoring for subgroup performance drift. This is the correct professional interpretation at final-report stage for responsible AI use in social-finance contexts.

Implementation and User Benefit

Evidence-to-Action Linkage

Validated Finding	Implementation Decision	Why It Improves Utility
Behavior lift is significant (RQ3)	Use behavior-rich scoring, not demographic-only triage	Higher discrimination supports better prioritization quality.
Top SHAP drivers are execution/hardship (RQ2)	Center interventions on cash-flow resilience and follow-through coaching	Targets highest-impact modifiable drivers.
Clusters are strongly separated (RQ4)	Route users by archetype-specific	Supports differential support intensity

	journeys	and content.
Subgroup recall gaps exist	Add threshold audits and periodic fairness-style monitoring	Reduces deployment risk from uneven error burden.

Table 21. Evidence-to-action implementation mapping.

Where the System Can Be Implemented

Primary deployment settings include financial counseling programs, employer wellness initiatives, credit counseling organizations, and fintech apps that collect periodic behavioral check-ins. The model can run in batch mode for monthly cohort screening or near-real-time mode for in-app intervention prompts.

Who Benefits

- End users: receive prioritized behavior-change guidance instead of generic advice.
- Counselors and educators: obtain interpretable risk tiers and key behavioral drivers for action planning.
- Program managers: gain segment-level dashboards for resource allocation and impact monitoring.
- Fintech product teams: can personalize nudges and workflows based on evidence-backed behavioral archetypes.

How the Operational Workflow Would Run

- Step 1: Collect user responses mapped to retained model fields (or mapped proxies).
- Step 2: Apply preprocessing rules consistent with the training pipeline (coding and imputation).
- Step 3: Score model probability for high-risk/low-well-being status.
- Step 4: Extract SHAP driver summary for each user record.
- Step 5: Assign cluster archetype and route intervention module.

- Step 6: Track user progress and re-score periodically to evaluate movement.

Example Intervention Mapping

Archetype	Observed Pattern	Recommended Intervention
Reactive-Strained	High hardship + low confidence + weak planning	Cash-flow stabilization, shock buffer setup, weekly follow-through coaching
Transitional Planner	Moderate planning with inconsistent execution	Goal decomposition, tracking cadence, accountability nudges
Stable Planner	Strong planning and low hardship	Maintenance nudges, long-horizon optimization guidance

Table 22. Example archetype-to-intervention mapping for operational use.

Ethics, Explainability, and Governance

This system should be deployed as decision support, not as an autonomous gatekeeping tool. Model outputs require human interpretation, and SHAP explanations should be surfaced to practitioners to avoid black-box decision risk. Sensitive group fairness checks and drift monitoring should be added before high-stakes production use.

Limitations and Further Improvements

Current Limitations

- Cross-sectional design does not establish causal relationships between behavior and outcome.
- Survey responses are self-reported and can include recall or desirability bias.
- Two requested fields ("TRACK", "PAYCHECK") are unavailable in the official public-use schema.
- Weighted analysis is currently strongest in descriptive robustness; full survey-design-aware model inference is future work.

- External validation on a different wave or independent dataset has not yet been completed.

Risk Mitigation Already Applied

- Codebook-driven handling of non-substantive codes to avoid hidden value contamination.
- Threshold sensitivity testing at 54 and 56 to reduce single-cutoff fragility.
- Multiple metric views (AUC, log-loss, Brier, F1) to avoid one-metric bias.
- DeLong tests and ANOVA/Tukey inference layers to strengthen statistical defensibility.
- Reproducible artifact pipeline for end-to-end traceability.

Future Scope

Planned Next Improvements

- Extend subgroup diagnostics to full demographic intersections and equalized-odds style checks.
- Implement survey-design-aware weighted model estimation and compare with current unweighted and weighted-context views.
- Evaluate monotonic constraints and simpler interpretable surrogate models for governance settings.
- Integrate post-intervention tracking to move from predictive analytics to longitudinal outcome-improvement analytics.
- Add external-wave validation when compatible CFPB follow-up data is available.

Conclusion

This capstone provides a complete, evidence-based demonstration that behavior-centered modeling substantially improves financial well-being classification and produces actionable explanations. Results are statistically strong, operationally interpretable, and reproducible from official source files. The project is therefore suitable for high-quality academic assessment and practical extension into intervention systems.

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Appendix

Appendix A: Repository Structure and Dataset Preview

Figure A1.

Repository folder structure prepared for reproducible submission.

Figure A1. GitHub repository folder structure

```

financial-wellbeing-capstone-github-final/
├── audit/
│   ├── cluster_summary.json
│   ├── csv_profile.json
│   ├── final_dataset_column_profile.csv
│   ├── final_dataset_summary.json
│   ├── final_dataset_variable_rationale.csv
│   ├── imputation_log.csv
│   ├── model_sanity_check.json
│   ├── official_source_verification.csv
│   ├── row_filter_log.csv
│   ├── variable_missingness_quality_report.csv
│   └── variable_selection_audit.csv
├── data/
│   ├── processed/
│   │   ├── analysis_dataset.csv
│   │   ├── clustered_analysis_dataset.csv
│   │   └── final_dataset_capstone_v2.csv
│   └── raw/
│       ├── cfpb_nfwbs-puf-codebook.pdf
│       ├── cfpb_nfwbs-puf-user-guide.pdf
│       └── NFWBS_PUF_2016_data.csv
├── docs/
│   ├── QM-640-Final-Synopsis.pdf
│   ├── QM-640-Interim-Report-draft.docx
│   └── QM-640-Interim-Report-draft.md
├── notebooks/
│   ├── 01_eda.ipynb
│   ├── 02_modelling_rq1_rq3.ipynb
│   ├── 03_clustering_rq4.ipynb
│   └── 04_shap_rq2.ipynb
├── outputs/
│   ├── figures/
│   │   ├── rq1_best_model_confusion_matrix.png
│   │   ├── rq1_roc_curves.png
│   │   ├── rq2_shap_beeswarm.png
│   │   ├── rq2_shap_dependence_propplan4_goalconf.png
│   │   ├── rq2_shap_top15_bar.png
│   │   ├── rq4_cluster_profile_heatmap.png
│   │   ├── rq4_hierarchical_dendrogram.png
│   │   └── rq4_kmeans_selection_metrics.png
│   └── tables/

```

Figure A2.

Preview of the final modeling dataset "final_dataset_capstone_v2.csv".

Figure A2. Preview of cleaned modeling dataset (first 12 rows)

FWBscore	FWB_high_54	FSscore	KHscore	PLAN_HORIZON	HARDSHIP_TOTAL	finalwt
55.0	1.0	44.0	1.267	3.0	0.0	0.367
51.0	0.0	43.0	-0.57	3.0	0.0	1.328
49.0	0.0	42.0	-0.188	4.0	0.0	0.835
49.0	0.0	42.0	-1.9	3.0	6.0	4.261
67.0	1.0	57.0	0.242	1.0	0.0	0.76
51.0	0.0	54.0	1.267	3.0	0.0	1.371
47.0	0.0	35.0	-0.909	2.0	4.0	1.193
43.0	0.0	58.0	-0.188	3.0	1.0	2.498
58.0	1.0	42.0	-0.57	4.0	0.0	1.267
78.0	1.0	66.0	-0.188	5.0	0.0	0.938
62.0	1.0	57.0	0.242	4.0	0.0	0.745
50.0	0.0	49.0	-0.909	1.0	1.0	1.293

Appendix B: Extended Data Dictionary (Selected)

Variable	Group	Missing %	Decision	Reason
FWBscore	outcome	0.08	keep	outcome variable
SAVEHABIT	behavioral	0.31	keep	missing <= 15%
SCFHORIZON	behavioral	1.28	keep	missing <= 15%
PROPLAN_1	behavioral	0.16	keep	missing <= 15%
PROPLAN_2	behavioral	0.16	keep	missing <= 15%
PROPLAN_3	behavioral	0.16	keep	missing <= 15%
PROPLAN_4	behavioral	0.16	keep	missing <= 15%
MANAGE1_1	behavioral	0.19	keep	missing <= 15%
MANAGE1_2	behavioral	0.19	keep	missing <= 15%
MANAGE1_3	behavioral	0.17	keep	missing <= 15%
MANAGE1_4	behavioral	0.17	keep	missing <= 15%
ACT1_1	behavioral	0.38	keep	missing <= 15%
ACT1_2	behavioral	0.28	keep	missing <= 15%
FINGOALS	behavioral	1.19	keep	missing <= 15%
GOALCONF	behavioral	0.44	keep	missing <= 15%
DISTRESS	stress_hardship	0.77	keep	missing <= 15%
ABSORBSHOCK	stress_hardship	0.58	keep	missing <= 15%
VOLATILITY	stress_hardship	1.00	keep	missing <= 15%
ENDSMEET	stress_hardship	0.69	keep	missing <= 15%
COVERCOSTS	stress_hardship	0.91	keep	missing <= 15%
MATHARDSHIP_1	stress_hardship	0.42	keep	missing <= 15%
MATHARDSHIP_2	stress_hardship	0.42	keep	missing <= 15%
MATHARDSHIP_3	stress_hardship	0.41	keep	missing <= 15%
MATHARDSHIP_4	stress_hardship	0.42	keep	missing <= 15%
MATHARDSHIP_5	stress_hardship	0.39	keep	missing <= 15%

Table A1. Extended data dictionary subset with selection rationale.

Appendix C: Additional EDA Visuals

Figure C1.

Correlation heatmap for selected continuous predictors.

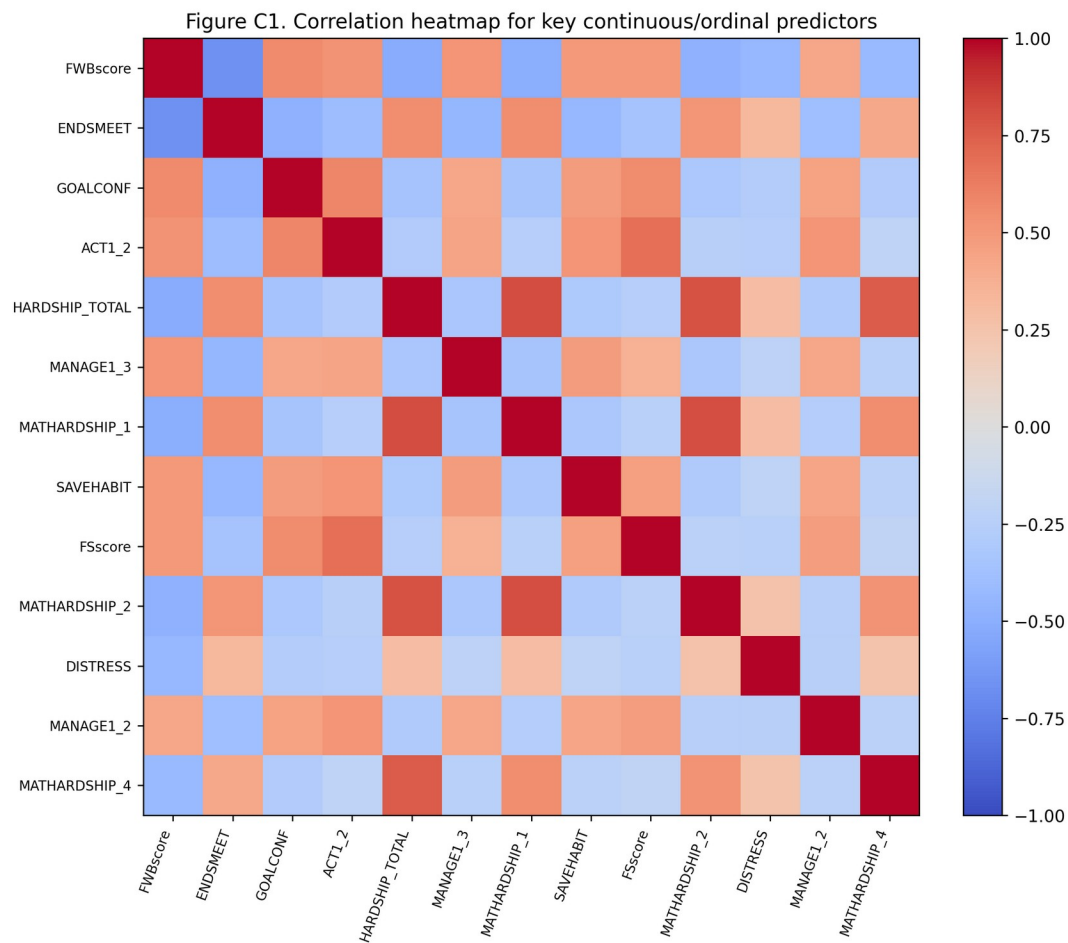


Figure C2.

Distribution of "FSscore" and "KHscore".

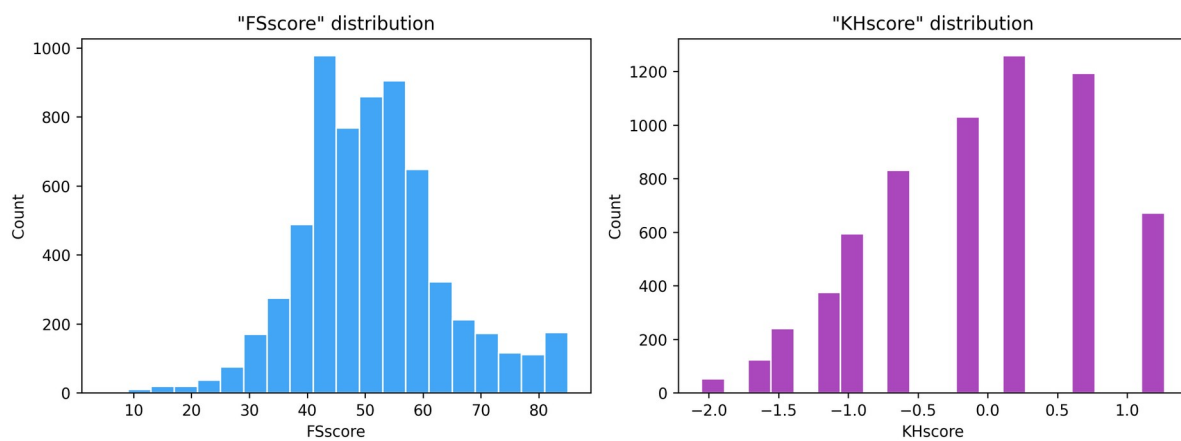


Figure C3.

"FWBscore" by income category.

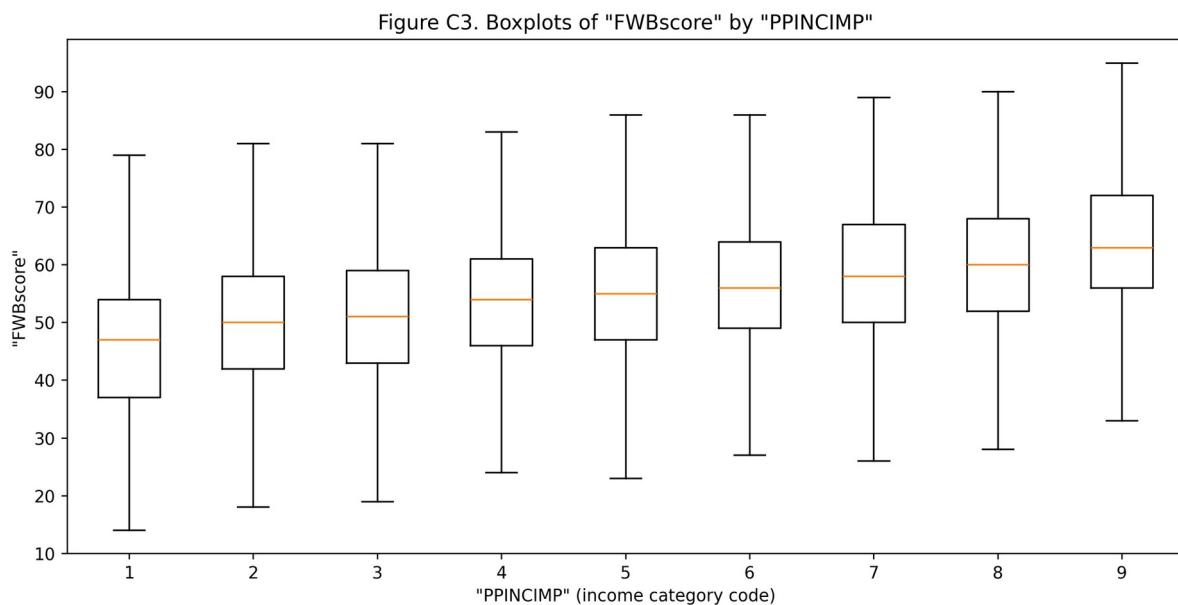
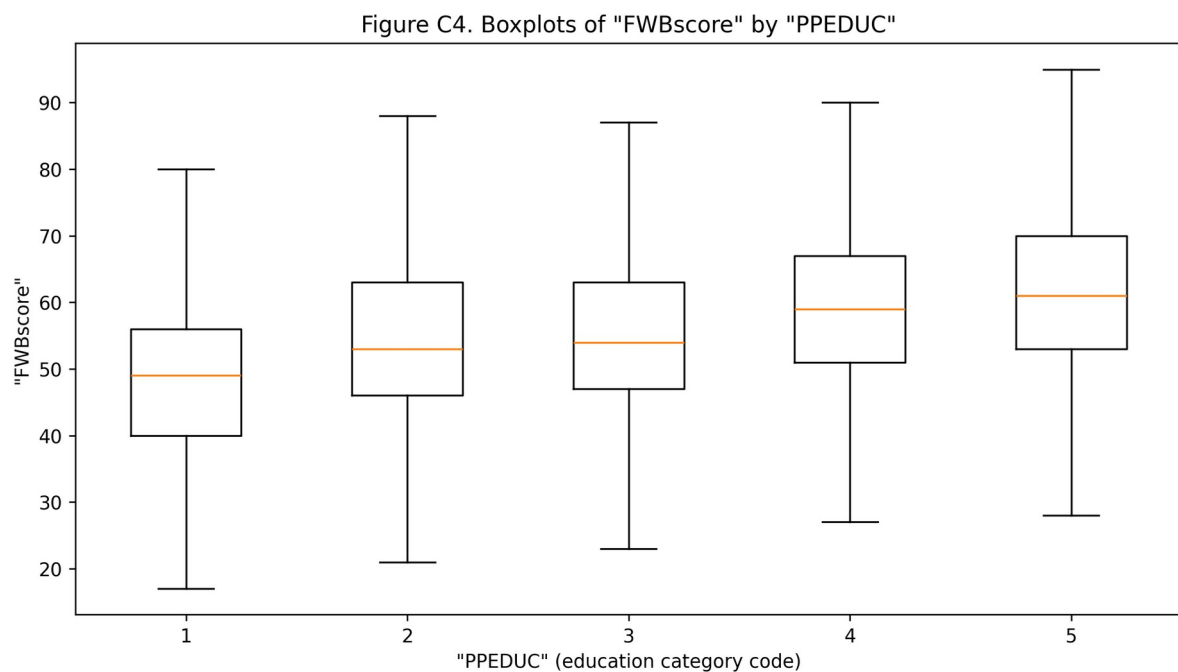


Figure C4.

"FWBscore" by education category.



Appendix D: Additional Inferential Tables

Group 1	Group 2	Mean Difference	Adjusted p	Reject H0
0	1	12.2464	0.000000	True
0	2	-12.0737	0.000000	True
1	2	-24.3201	0.000000	True

Table A2. Tukey post-hoc comparisons for cluster mean differences.

Source	Sum Sq	df	F	p-value
C(behavior_cluster_final)	248521.568	2.0	1146.791	0
agecat	16859.518	1.0	155.595	2.67779e-35
PPINCIMP	37395.455	1.0	345.119	4.59864e-75
PPEDUC	695.795	1.0	6.421	0.011299
EMPLOY	4478.779	1.0	41.334	1.37608e-10
PPGENDER	404.557	1.0	3.734	0.0533722
PPETHM	120.420	1.0	1.111	0.29183
Residual	689680.890	6365.0		

Table A3. Controlled ANOVA with demographic covariates.

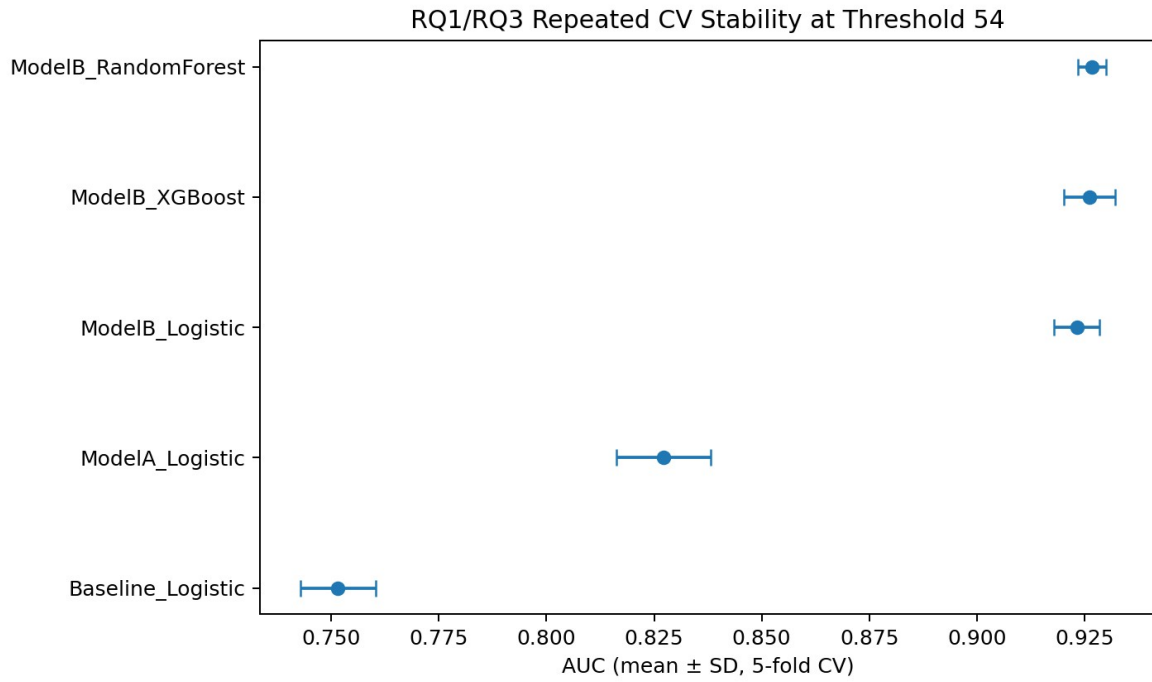
Appendix E: Advanced Validation Addendum

Model	CV AUC Mean	CV AUC SD	CV F1 Mean	CV Log-loss Mean
ModelB_RandomForest	0.9267	0.0033	0.8742	0.3463
ModelB_XGBoost	0.9262	0.0059	0.8751	0.3443
ModelB_Logistic	0.9233	0.0053	0.8728	0.3521
ModelA_Logistic	0.8272	0.0109	0.7912	0.5080
Baseline_Logistic	0.7516	0.0087	0.7489	0.5844

Table A4. Repeated cross-validation summary (appendix view).

Figure E1.

Repeated CV AUC error-bar plot.



Variant	Features	AUC	Log-loss	F1
ModelB_Full	61	0.9326	0.3291	0.8732
ModelB_Ablated_No_Proximal_StressHardship	49	0.9006	0.3942	0.8487
Delta_Full_minus_Ablated	12	0.0320	-0.0651	0.0245

Table A5. Proximal-feature ablation summary.

Dimension	Metric	Max	Min	Gap
age_group_3	auc_roc	0.9359	0.8354	0.1005
age_group_3	recall	0.9211	0.6087	0.3124
age_group_3	false_negative_rate	0.3913	0.0789	0.3124
education_group_3	auc_roc	0.9544	0.9150	0.0394
education_group_3	recall	0.9379	0.8081	0.1299
education_group_3	false_negative_rate	0.1919	0.0621	0.1299
income_group_3	auc_roc	0.9330	0.9100	0.0230
income_group_3	recall	0.9472	0.7893	0.1580
income_group_3	false_negative_rate	0.2107	0.0528	0.1580

Table A6. Subgroup gap summary for key metrics.

Appendix F: Canonical Evidence Freeze

Metric Key	Metric Value
dataset_raw_rows	6394
dataset_raw_cols	217
dataset_final_rows	6374
dataset_final_cols	70
dataset_retained_source_vars	64
dataset_derived_vars	6
rq1_auc_baseline_54	0.758
rq1_auc_modelA_54	0.8294

rq1_auc_modelB_xgb_54	0.9329
rq1_logloss_modelB_xgb_54	0.3288
rq1_f1_modelB_xgb_54	0.8812
rq1_auc_modelB_xgb_56	0.9199
rq3_delong_z_54	10.4983
rq3_delong_p_54	0.0
rq3_delong_z_56	8.8188
rq3_delong_p_56	0.0
rq2_top_shap_1_feature	ENDSMEET
rq2_top_shap_1_value	1.0428
rq2_top_shap_2_feature	ABSORBSHOCK
rq2_top_shap_2_value	0.4037
rq4_best_k	3
rq4_anova_f	1964.9096
rq4_anova_p	0.0
rq4_eta_squared	0.3815
weighted_mean_fwb	54.2403
weighted_high54_rate	0.5194
weighted_high56_rate	0.4604
cv_best_model	ModelB_RandomForest
cv_best_auc_mean	0.9267
cv_best_auc_sd	0.0033
ablation_auc_drop	0.032
subgroup_max_recall_gap	0.3124
rq3_auc_delta_54	0.0988
rq3_auc_delta_54_ci_low	0.0812
rq3_auc_delta_54_ci_high	0.1178
rq3_auc_delta_56	0.081
rq3_auc_delta_56_ci_low	0.0636
rq3_auc_delta_56_ci_high	0.0988
rq4_cohen_f	0.7854
rq4_cramers_v	0.5787
inference_alpha	0.05
inference_target_power	0.8
inference_confidence_level	0.95
power_required_n_rq1_epv10	1440.0519
power_required_n_rq3_threshold54	90.7994
power_required_n_rq3_threshold56	128.6778
power_required_n_rq4_anova	18.9614

Table A7. Canonical numbers ledger used for final report/deck consistency.

Cluster	Final Label	N	Mean FWB	High54	High56
0	Transitional Planner	2104	51.14	0.376	0.295
1	Stable Planner	3414	63.39	0.827	0.771
2	Reactive-Strained	856	39.07	0.068	0.047

Table A8. Canonical final cluster labels for narrative consistency.